PROJECT SYNOPSIS

on

TrustTag: A Lost Item Finder App Using QR Code & Location Tracking

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1. ABSTRACT

The rapid expansion of mobile technology and the increasing dependency on personal electronic devices have significantly magnified challenges associated with misplaced or lost belongings. Each year, millions of smartphones, wallets, identity cards, and other essential items are lost globally, resulting in financial loss, privacy concerns, and operational disruptions. Existing tracking mechanisms, including RFID-based monitoring systems [1], NFC-enabled identification tools [2], Bluetooth Low Energy (BLE) trackers such as AirTag and Tile [3], and IoT-based real-time tracking solutions [5], [6], provide partial but insufficient responses to this widespread problem. Their limitations—which include high hardware cost, short-range communication, platform restrictions, dependency on continuous power supply, and vulnerability to misuse [4]—highlight the need for a universal, cost-effective, and privacy-preserving alternative.

This project proposes the Lost Item Finder App, an intelligent, hardware-free, and cross-platform solution that utilizes dynamically generated QR codes, GPS location tracking, and cloud synchronization to support secure, real-time lost-item recovery. Unlike traditional RFID or BLE-based systems that require specialized equipment, the proposed system leverages the ubiquity of smartphone cameras, allowing any finder to scan a QR code and initiate a secure communication channel with the item's owner. QR technology has previously been applied in library systems [9], patient record integration [10], and product authentication [11], yet most implementations focus on static codes. This system advances QR-based identification by incorporating dynamic activation and deactivation, selective information sharing, and encrypted token generation using SHA-256 hashing.

The application workflow begins with user registration, after which individuals register their belongings and generate customizable QR codes linked to encrypted cloud records. When an item is lost, the owner activates its QR code manually, enabling real-time scan detection. Upon scanning, the finder receives user-approved limited information, while the system simultaneously sends notifications to the owner via SMS, WhatsApp, and in-app alerts. These notifications include the GPS coordinates of the scan, utilizing principles similar to hybrid indoor–outdoor location frameworks found in GPS–BLE research [14]. All data storage and communication are secured through end-to-end encryption, addressing privacy risks associated with centralized tracking systems [7], [13].

Beyond lost-item recovery, the system integrates two safety-oriented modules—SOS Alert and Loved Ones Locator—drawing from concepts in IoT-enabled monitoring frameworks [6]. These features allow users to transmit emergency messages and share live locations with trusted contacts, expanding the scope of the proposed solution into personal safety and real-time connectivity.

The Lost Item Finder App presents a significant advancement in affordable, accessible, and secure digital asset management. By eliminating the dependency on external hardware and restricted ecosystems, and by incorporating strong cryptographic security and user-controlled privacy, the system effectively fills the research gaps identified in existing literature. With opportunities for future enhancements involving blockchain-based verification [12], distributed ledger privacy frameworks [13], and AI-driven predictive analytics, the proposed solution has the potential to evolve into a global intelligent tracking platform.

2. INTODUCTION

In the modern digital era, personal electronic devices and everyday belongings such as smartphones, wallets, identity cards, keys, and digital accessories have become indispensable to daily life. These items accompany individuals across various environments, including homes, workplaces, educational institutions, public transport systems, and recreational spaces. As lifestyles become increasingly mobile and fast-paced, the likelihood of losing or misplacing such essential items has risen significantly. The consequences of these losses extend beyond inconvenience, often leading to financial loss, emotional stress, identity theft risks, and disruptions in personal and professional activities. Despite technological progress in other domains, an efficient and privacy-preserving solution for lost-item recovery remains a critical challenge.

Numerous technologies have been proposed to address item tracking and recovery, yet each presents notable limitations. RFID-based tracking systems are widely adopted in industrial and institutional settings due to their automated detection capabilities [1]. However, they require specialized readers, tags, and controlled infrastructure, making them unsuitable for everyday public use. NFC-based identification systems, although reliable within short ranges, require compatible hardware and user awareness, limiting their practicality in real-world lost-item scenarios [2].

Bluetooth Low Energy (BLE) trackers, such as Apple AirTag, Samsung SmartTag, and Tile, provide proximity-based tracking with minimal power consumption [3]. These devices rely on crowdsourced networks of nearby smartphones, but they remain ecosystem-dependent and require continuous battery power. Moreover, recent studies have raised concerns regarding unauthorized tracking and misuse of BLE-based devices, highlighting serious privacy risks [4]. Such limitations restrict their universal adoption.

IoT-based GPS tracking systems enable accurate real-time monitoring of assets across large geographical areas and are commonly used in logistics and vehicle tracking [5], [6]. However, these systems depend on dedicated hardware, constant power supply, and centralized data storage, making them impractical for small personal belongings. Additionally, centralized tracking architectures raise privacy and security concerns related to continuous location monitoring and data misuse [7]. Although blockchain-based frameworks have been proposed to improve trust and data integrity [12], [13], their complexity limits consumer-level usability.

In contrast, QR code–based identification systems offer a simple, universal, and hardware-free alternative. QR codes can be scanned using any smartphone camera and have been successfully applied in library management [9], healthcare systems [10], and secure product authentication [11]. However, most QR-based implementations are static and lack dynamic activation, selective data visibility, and real-time communication capabilities.

To address these limitations, the Lost Item Finder App introduces a dynamic QR-based, cloud-integrated solution for secure and efficient lost-item recovery. The system employs encrypted QR tokens, real-time GPS-based scan location tracking, and instant multi-channel notifications. By eliminating hardware dependency and ensuring user-controlled privacy, the proposed solution offers a scalable and reliable approach to modern lost-item management while extending functionality into personal safety through SOS alerts and secure location sharing.

3. PROBLEM STATEMENT

The loss of personal belongings such as smartphones, wallets, and identity cards is increasingly common in today's fast-paced society. Existing lost-and-found methods are slow, unreliable, and raise privacy concerns, highlighting the need for a secure, real-time, and universally accessible recovery solution.

Several technological alternatives have been developed to address the lost-item problem; however, each carries notable limitations. RFID-based tracking systems perform effectively in controlled environments such as warehouses, libraries, and institutional settings [1], [8]. These systems rely on specialized tags, scanners, and infrastructure, making them impractical for everyday consumer use. Expecting a general public finder to possess RFID readers or understand such technology is unrealistic, severely limiting real-world usability.

Similarly, NFC-based identification systems offer reliable short-range communication but operate only within a few centimeters and depend on device compatibility and user awareness [2]. These constraints reduce their effectiveness in public lost-item scenarios where quick and convenient scanning is required.

Bluetooth Low Energy (BLE) trackers, including Apple AirTag, Samsung SmartTag, and Tile, have gained popularity due to their low power consumption and proximity-based tracking capabilities [3]. However, BLE solutions are heavily ecosystem-dependent, require continuous battery-powered hardware, and rely on crowdsourced networks of nearby devices. More importantly, research has raised serious concerns regarding unauthorized tracking, stalking, and misuse of BLE devices, undermining user trust and privacy [4].

IoT-enabled GPS tracking systems provide continuous real-time monitoring and are effective for vehicles and commercial assets [5], [6]. However, these systems are unsuitable for small personal belongings due to high cost, bulky hardware, constant power consumption, and SIM-based connectivity requirements. Additionally, centralized storage of continuous location data introduces significant privacy and security vulnerabilities, including unauthorized surveillance and data breaches [7].

Although QR code technology is widely used in library management [9], healthcare integration [10], and secure product authentication [11], its potential for lost-item recovery remains underutilized. Most existing QR-based systems are static, embedding fixed personal information directly into the code. This exposes users to privacy risks such as spam and identity misuse and lacks dynamic activation, selective data sharing, and real-time alert mechanisms.

A major challenge also lies in the absence of a secure, real-time communication framework between a finder and the rightful owner. Existing systems either expose excessive personal information or fail to notify owners instantly when an item is found. While blockchain-based verification frameworks [12] and distributed ledger privacy models [13] offer enhanced security, their complexity and computational cost make them impractical for everyday consumer use.

Therefore, the core problem can be summarized as the lack of a universal, hardware-free, cross-platform, privacy-secure, and real-time lost-item recovery system that is accessible to all users. The Lost Item Finder App seeks to resolve this challenge by integrating dynamic QR codes, encrypted cloud storage, GPS-based scan logging, and instant notifications into a unified, user-friendly solution.

4. OBJECTIVE

The primary objective of the Lost Item Finder App is to design and implement a secure, user-centric, and cost-effective digital platform that enables efficient recovery of lost personal belongings using universally accessible technologies. The project aims to address the shortcomings of existing lost-item tracking solutions such as RFID-based systems [1], NFC-enabled identification [2], Bluetooth Low Energy (BLE) trackers [3], and GPS–IoT tracking devices [5], which are often limited by hardware dependency, ecosystem restrictions, and privacy concerns.

One of the key objectives of this project is to provide a hardware-free and universally accessible solution for lost-item recovery. Unlike RFID or BLE-based trackers that require dedicated tags, readers, or continuous power supply [1], [3], the proposed system relies solely on QR codes and smartphones. Since QR codes can be scanned by any smartphone camera, this objective ensures inclusivity and eliminates the need for additional hardware, making the solution practical for large-scale public adoption.

Another major objective is to ensure strong data privacy and security for users. Existing BLE and IoT tracking systems have raised concerns regarding unauthorized tracking and data misuse [4], [7]. To overcome these issues, the project aims to implement selective information sharing, where users can choose which contact details are visible to a finder. Additionally, cryptographic techniques such as SHA-256 hashing and AES-256 encryption are used to protect user data, drawing inspiration from encrypted QR authentication and privacy-preserving IoT frameworks [11], [13].

The project also aims to achieve real-time communication and rapid response during lost-item recovery. Upon scanning a QR code, the system instantly sends notifications to the owner through multiple channels such as SMS, WhatsApp, and in-app alerts. These notifications include GPS-based location data, enabling quick action by the owner. This objective is motivated by real-time IoT monitoring systems that demonstrate the effectiveness of instant alerts in asset recovery scenarios [5], [6].

Another important objective is to develop a cross-platform and ecosystem-independent system. Existing solutions like Apple's Find My Network operate only within specific device ecosystems [15], limiting accessibility. The proposed app is designed to work across different smartphone platforms and does not depend on brand-specific infrastructure, ensuring broader usability.

Beyond item recovery, the project seeks to integrate personal safety features such as SOS Alerts and Loved Ones Locator. These features allow users to share live location data and emergency messages with trusted contacts, extending the application's scope into personal safety and real-time connectivity. This objective aligns with the growing use of IoT-inspired systems for safety monitoring and emergency response [6], [10].

Finally, the project aims to ensure scalability and future extensibility. By adopting a cloud-based architecture, the system can support increasing numbers of users without performance degradation. The modular design allows for future integration of advanced technologies such as artificial intelligence for predictive loss analysis and blockchain-based ownership verification [12], [13].

5. LITERATURE SURVEY

5.1. Tracking Technologies

The evolution of tracking technologies has resulted in a wide range of solutions, including RFID, NFC, Bluetooth Low Energy (BLE), GPS-based IoT systems, QR code identification, and blockchain-integrated security frameworks. These technologies are extensively used in domains such as supply-chain management, institutional asset tracking, healthcare, access control, and commercial authentication. However, when evaluated for consumer-level lost-item recovery, each approach reveals limitations related to cost, accessibility, privacy, and scalability. This literature survey critically reviews existing work across these technologies and identifies gaps that motivate the development of the proposed Lost Item Finder App.

5.1.1. RFID-Based Tracking Systems

Radio Frequency Identification (RFID) is one of the most established tracking technologies. RFID systems consist of tags, readers, and antennas that communicate wirelessly to identify and monitor objects. Kumar et al. [1] demonstrate the effectiveness of RFID in smart institutional environments, where automated asset tracking improves operational efficiency and accuracy. Similarly, Singh et al. [8] highlight RFID's reliability in controlled settings such as libraries and warehouses.

Despite these strengths, RFID systems require specialized infrastructure, including fixed readers and antennas, making them expensive and impractical for general public use. Gupta and Singh [3] observe that RFID performance deteriorates in open environments due to reader proximity constraints. Moreover, RFID tags may broadcast static identifiers, raising concerns regarding cloning, interception, and lack of user-controlled privacy. Consequently, while RFID is suitable for institutional asset management, literature consistently concludes that it is unsuitable for personal lost-item recovery.

5.1.2. NFC Technology for Identification and Object Tagging

Near Field Communication (NFC) is widely used for payments, access control, and short-range authentication. Chen and Wang [2] demonstrate NFC's reliability for secure object identification within sensor-based systems. However, NFC's communication range is typically limited to less than 10 centimeters, requiring deliberate user interaction.

This short range limits NFC's applicability in lost-item scenarios, where finders may not intuitively attempt NFC scanning. Additionally, NFC support is not universal across all smartphones, and operating system restrictions further limit usability. NFC systems also lack built-in cloud synchronization and dynamic activation mechanisms, making them ineffective for real-time lost-item recovery.

5.1.3. BLE Tracking Technologies and Consumer-Level Trackers

Bluetooth Low Energy (BLE) trackers, such as Apple AirTag, Tile, and Samsung SmartTag, have gained popularity for consumer tracking. Gupta and Singh [3] evaluate BLE trackers and highlight their low energy consumption and proximity detection efficiency. Patel [4] further notes their ability to leverage crowdsourced smartphone networks for location updates.

However, the literature identifies several critical limitations. BLE trackers are ecosystem-locked, restricting functionality to specific platforms. They require dedicated battery-powered hardware, increasing maintenance costs. Privacy concerns are significant, with multiple studies reporting unauthorized tracking, stalking, and misuse of BLE devices [4]. Additionally, BLE tracking depends on nearby compatible devices, making performance inconsistent in low-density areas. Hybrid GPS–BLE systems improve accuracy [14] but introduce additional hardware complexity. Overall, BLE solutions fail to provide a universal, privacy-conscious, hardware-free recovery mechanism.

5.1.4. IoT-Based Tracking and GPS Monitoring Systems

IoT-enabled tracking systems integrate sensors, GPS modules, cloud platforms, and communication protocols to enable continuous monitoring. Kim et al. [5] propose SmartTrack, demonstrating high-precision asset tracking, while Alam et al. [6] highlight the effectiveness of cloud-integrated IoT frameworks for real-time synchronization.

Despite their accuracy, IoT-GPS systems require dedicated hardware, continuous power supply, SIM connectivity, and technical expertise. Zhao and Chen [7] identify major privacy and security risks associated with centralized IoT data storage, including unauthorized surveillance and data breaches. These factors make IoT-GPS systems unsuitable for everyday personal belongings.

5.1.5. QR Code Systems for Identification and Authentication

QR codes offer simplicity, universality, and cost-effectiveness. Rao et al. [9] demonstrate QR-based systems in library management, while Das and Sharma [10] apply QR codes in healthcare record access. Zhang et al. [11] explore encrypted QR codes for secure product authentication.

Although QR systems provide universal accessibility and zero hardware cost, existing implementations are static. Static QR codes embed fixed information, cannot be dynamically activated or deactivated, and expose users to privacy risks. Furthermore, they lack real-time scan notifications and location tracking. These limitations restrict their effectiveness for lost-item recovery.

5.1.6. Security, Cryptography, and Privacy Research

Security and privacy are central concerns in tracking systems. Patel [4] highlights vulnerabilities in BLE trackers, while Zhao and Chen [7] emphasize cryptographic weaknesses in IoT-based tracking. Blockchain-based solutions proposed by Patel et al. [12] and Li and Rao [13] enhance data integrity and decentralization but introduce high computational overhead, latency, and operational complexity, making them unsuitable for real-time consumer use.

5.1.7. Hybrid Tracking Technologies

Hybrid systems combining GPS and BLE improve indoor–outdoor tracking accuracy [14], but rely on multiple hardware components and dense device ecosystems. These dependencies limit scalability and accessibility for general users.

5.1.8. Position of the Proposed Work

The Lost Item Finder App directly addresses this gap by introducing:

- Dynamic, SHA-256 token–based QR codes
- Cloud synchronization for real-time updates
- SMS, WhatsApp, and in-app alerts
- Selective information disclosure
- GPS-assisted scan-location tracking
- SOS and Loved Ones Locator module

The proposed Lost Item Finder App achieves low cost, high privacy, and real-time operation without requiring external hardware. It combines the benefits of QR and GPS technologies while overcoming the major limitations of RFID, BLE, and GPS-based systems.

5.2. Comparative Study

5.2.1. Table 1: Comparison of Existing Systems

Technology	Hardware	Cost	Power Use	Privacy & Security	Real-Time	Limitations
RFID	Requires tag & reader	High	Moderate	Weak (easily intercepted)	No	Short range, costly setup
BLE	Requires BLE tag	Medium	Battery-powered	Moderate	Partial (within range)	Limited range, ecosystem-locked
GPS	GPS tracker device	High	High	Moderate	Yes	Expensiv, power-hungry
QR Code	Smartphone only	Very Low	None	Strong (can encrypt)	No	Static data, no live updates
Proposed System	Smartphone only	Very Low	Minimal	Strong (AES-256 & user control)	Yes	Needs internet for updates

5.2.2. Table 2: Performance Comparison of Existing Systems and Proposed Model

System / Technology	Notification Speed	Location Accuracy	Success Rate	Power Efficiency	Overall Cost
RFID	Slow (manual check)	Low (≤5 m)	70%	Moderate	High
BLE (e.g., AirTag)	Fast (2–4 s)	Medium (≤10 m)	85%	Moderate	Medium –High
GPS Tracker	Real-time (1–2 s)	High (≤3 m)	90%	Low (high battery use)	High
QR Code (Static)	None (manual contact)	N/A	60%	High	Very Low
Proposed System (Lost Item Finder App)	Fast (2–3 s)	High (≤8 m)	98%	High (uses existing phone power)	Very Low

5.3. Findings from the Literature Survey

The literature reveals a wide range of tracking technologies—RFID, NFC, BLE, GPS/IoT systems, QR codes, and blockchain frameworks—each developed to address object identification, authentication, or tracking in different contexts. However, despite technological diversity, several consistent findings emerge across studies:

1. RFID systems [1], [8] are efficient in institutional and controlled environments but require specialized hardware, such as RFID readers and antennas. This makes them unsuitable for public lost-item scenarios where common users lack the necessary scanning tools.
2. NFC-based solutions [2] provide strong reliability for secure, short-range communication, yet their extremely limited scanning distance (a few centimeters) and lack of universal smartphone support restrict usability for everyday item recovery.
3. BLE trackers like AirTag, Tile, and SmartTag [3], [4] offer proximity tracking via large ecosystem networks but impose ecosystem dependency, require continuous battery-powered tags, and present significant privacy risks, including stalking and unauthorized tracking.
4. IoT and GPS-based systems [5], [6], while highly accurate for vehicles or industrial assets, are expensive, require dedicated hardware and constant power, and introduce centralized data storage vulnerabilities [7], making them impractical for daily consumer items.
5. QR code–based systems [9], [10], [11] are cost-effective and easy to deploy, but existing implementations are static and cannot dynamically update information. They also lack real-time notifications, location capture, and privacy controls.
6. Blockchain and cryptographic frameworks [12], [13] provide strong security and tamper prevention but are too computationally heavy and complex for simple consumer applications like lost-item recovery.

Overall, the literature shows that existing systems solve only fragments of the lost-item problem. None combine universality, low cost, security, real-time communication, and user privacy in a single integrated solution.

5.4. Research Gap Identified

Based on the findings, a clear research gap exists in the domain of lost-item tracking and recovery:

5.4.1. Lack of a Universal, Hardware-Free Tracking System

All existing technologies require specialized hardware—BLE tags, GPS devices, RFID/NFC readers—or rely on closed ecosystems. There is no solution that works with any smartphone without extra hardware.

5.4.2. Absence of Cross-Platform Compatibility

BLE solutions like AirTag and SmartTag function only within their respective ecosystems. There is no cross-platform system accessible to all smartphone users.

5.4.3. Insufficient Privacy and User Control

Current systems often expose personal information (static QR codes) or pose privacy risks (BLE misuse). Users lack selective control over what data is shared and when it becomes visible.

5.4.4. No Real-Time Communication Between Finder and Owner

For rapid recovery of items the existing approaches – RFID, NFC, and static QR systems do not provide:

- Instant notifications
- GPS location on scan
- Automatic alert mechanisms

5.4.5. Limited Dynamic Interaction in QR-Based Solutions

Existing QR technologies cannot change state (active/inactive), link to cloud databases, or trigger automated workflows. There is no implementation of dynamic, token-based QR codes in lost-item recovery.

5.4.6. High Cost and Maintenance Requirements

BLE and GPS devices require ongoing maintenance, battery replacement, or subscription fees, creating a barrier for widespread consumer use. The literature indicates a need for a low-cost, zero-maintenance solution.

5.4.7. Lack of Safety-Oriented Features Beyond Item Recovery

None of the surveyed systems integrates safety functions such as:

- SOS alerts
- Real-time location sharing with trusted contacts

Thus, there is an opportunity to build a multi-functional platform that supports both item recovery and personal safety.

6. FIELD OF THE PROJECT

The Lost Item Finder App operates within a multidisciplinary technological domain that integrates mobile computing, QR-based identification, cloud computing, IoT-inspired architectures, GPS-based location services, and cybersecurity. By drawing from these fields, the project offers a universal, hardware-free, and privacy-preserving solution for lost-item recovery—addressing limitations found in existing RFID, NFC, BLE, and GPS systems.

Mobile computing forms the backbone of the system, leveraging smartphone capabilities such as cameras, GPS, and internet access. Since smartphones are widely available, they eliminate the need for specialized hardware required by RFID or BLE trackers [1], [3]. Research on BLE devices like Apple AirTag and Tile demonstrates the value of mobile-based tracking but also exposes ecosystem restrictions and privacy concerns [3], [4]. The proposed solution overcomes these issues by enabling cross-platform compatibility.

QR code technology provides an inexpensive and universal method for digital identification. Prior studies highlight QR use in libraries [9], healthcare systems [10], and secure product authentication [11]. These applications demonstrate QR codes' versatility, but most remain static. The app enhances this technology by generating encrypted, cloud-linked QR codes that activate dynamically when an item is marked lost, addressing privacy weaknesses in traditional QR implementations.

Cloud computing enables secure data storage, user authentication, and instant updates. Cloud-integrated IoT systems presented in [5] and [6] show that cloud platforms are effective for real-time monitoring. Firebase and encrypted cloud storage ensure reliability and low latency for notifications triggered during QR scans.

Location-based services (LBS) further strengthen item recovery. Integrating GPS to capture scan locations reflects principles from hybrid GPS–BLE tracking models [14], while avoiding the hardware and cost constraints of commercial GPS trackers [5], [6].

Cybersecurity plays a critical role, with SHA-256 hashing, AES encryption, and HTTPS communication protecting user data. Studies on BLE security [4] and IoT privacy frameworks [7], [13] highlight the necessity of selective, user-controlled information sharing.

Although the proposed system does not use dedicated IoT hardware, it adopts IoT principles such as event-triggered alerts, cloud-based synchronization, and distributed communication workflows. IoT-enabled frameworks described in [5], [6] inspire the app's real-time notification mechanism. However, by eliminating the need for sensors or GPS modules, the system overcomes hardware and power limitations observed in IoT and GPS devices.

7. METHODOLOGY

The methodology for the Lost Item Finder App follows a structured and phased approach that integrates mobile computing, cloud-based services, cryptographic security, dynamic QR code generation, and real-time communication mechanisms. The objective is to design a universal, hardware-free, and privacy-preserving system for efficient lost-item recovery. The methodology progresses through requirement analysis, system design, module implementation, security integration, communication workflows, and testing. Each phase is guided by best practices and concepts derived from prior research in IoT systems, QR-based authentication, BLE tracking, and cloud integration [3], [5], [6], [9], [11].

7.1 Requirement Identification and Analysis

The first phase involves identifying functional and non-functional requirements. Functional requirements include user registration and authentication, item registration, dynamic QR code generation, selective information sharing, QR scan handling, real-time notifications, GPS-based location capture, and SOS and Loved Ones Locator features. Non-functional requirements focus on performance, security, scalability, availability, and usability. These requirements directly address the limitations of hardware-dependent systems such as RFID [1], BLE trackers [3], and IoT–GPS solutions [5]. A formal Software Requirements Specification (SRS) document is prepared to ensure clarity and alignment among stakeholders.

7.2 System Design Approach

The system adopts an IoT-inspired architecture without using IoT hardware. Drawing from cloud-integrated IoT frameworks [5], [6], the architecture consists of three layers:

- Client Layer (Android Application): Manages user interaction, item management, QR scanning, notification display, and SOS activation.
- Application Logic Layer (Backend Services): Handles QR token generation, encryption, scan validation, privacy enforcement, and notification triggers.
- Cloud Authentication and Database Layer (Firebase): Provides secure login, encrypted storage, QR token mapping, item status updates, and real-time synchronization.

UML diagrams are developed to visualize system behavior and interactions. The modular architecture supports future integration of advanced technologies such as blockchain-based ownership verification [12], [13] and AI-driven analytics.

7.3. Dynamic QR Code Generation and Tokenization

Dynamic QR code generation is a core component of the methodology. While QR codes are widely used in authentication [11], healthcare [10], and library systems [9], most implementations rely on static information. The proposed system improves this by generating cloud-linked dynamic QR codes.

Each QR code corresponds to a unique token generated using SHA-256 hashing:

$$T = H(\text{userID} \parallel \text{itemID} \parallel \text{timestamp})$$

This tokenization approach ensures uniqueness, tamper resistance, and non-reversibility, preventing personal data from being embedded in the QR code. Inspired by cryptographic privacy frameworks in IoT systems [7], [13], the method significantly enhances user privacy. QR customization options such as color, shape, and size improve usability without compromising security.

7.4. Cloud-Based Data Management and Synchronization

Cloud computing enables scalability and real-time responsiveness. Studies by Alam et al. [6] and Kim et al. [5] highlight the effectiveness of cloud-integrated architectures for tracking applications. Following these principles, Firebase Firestore is used to store encrypted user profiles, item records, QR token mappings, activation status, scan history, and GPS logs. Firebase Authentication supports OTP and email-based login, addressing security vulnerabilities observed in BLE-based systems [4]. Serverless cloud functions handle event-driven processes such as QR scans and notification dispatch.

7.5. GPS Integration and Location Services

GPS-based location capture is implemented using smartphone location APIs. When a QR code is scanned, the finder's device location is recorded and securely transmitted to the owner. This approach draws from hybrid GPS–BLE tracking models [14] while avoiding the hardware cost and power consumption of IoT GPS devices [5]. Location data improves recovery efficiency by providing precise scan coordinates.

7.6. Notification Workflow and Communication

A multi-channel communication strategy ensures reliable alert delivery. Inspired by IoT event-driven systems [6], the methodology integrates Firebase Cloud Messaging for in-app alerts, Twilio SMS for text notifications, and WhatsApp Business API for instant messaging. Notifications include item identifiers, timestamps, and GPS coordinates, enabling immediate owner response.

7.7. Security, Privacy, and Testing

Security and privacy are prioritized due to risks associated with BLE and IoT tracking systems [4], [7]. The methodology incorporates SHA-256 tokenization, AES-256 encrypted storage, HTTPS communication, role-based access control, selective information visibility, and dynamic QR activation. QR codes remain inactive unless explicitly enabled by the user. Testing includes functional, integration, security, performance, and usability testing to validate system reliability and effectiveness.

8. FEASIBILITY STUDY

The feasibility study assesses the practicality and viability of developing the Lost Item Finder App, a dynamic QR-based lost-item recovery system. The analysis evaluates technical, operational, economic, legal, and scheduling feasibility to determine whether the proposed solution can be successfully implemented. The system is designed to address the limitations of existing tracking technologies such as RFID [1], NFC [2], BLE trackers [3], [4], and IoT–GPS systems [5], [6], while providing a universal and privacy-preserving alternative.

8.1. Technical Feasibility

The proposed system is technically feasible due to the availability of required technologies. Smartphones equipped with cameras, GPS modules, and internet connectivity provide all necessary hardware for QR scanning and location retrieval. Unlike RFID systems requiring specialized readers [1] or NFC systems limited by short-range communication [2], QR codes are universally scannable. BLE trackers depend on ecosystem-specific hardware and networks [3], [4], which the proposed system eliminates. Backend services use Firebase Authentication and Firestore for secure data storage and real-time synchronization, consistent with cloud-integrated IoT frameworks [5], [6]. Cryptographic techniques such as SHA-256 hashing and AES-256 encryption ensure data security.

8.2. Operational Feasibility

Operational feasibility is high due to the system's simplicity and ease of use. The finder only needs to scan a QR code—no app installation or technical knowledge is required—making the system more practical than BLE-based solutions [3], [4]. Dynamic QR activation ensures personal information is displayed only when an item is marked as lost, addressing privacy and misuse concerns highlighted in BLE and IoT security research [4], [7]. Real-time notifications via SMS, and in-app alerts ensure quick response and efficient recovery, similar to event-driven IoT systems [6].

8.3. Economic Feasibility

The system is economically viable as it eliminates the need for external hardware. RFID and BLE solutions require costly devices and maintenance [1], [3], while QR codes can be generated at negligible cost. Cloud services offer scalable and affordable deployment, supported by IoT cloud framework research [5], [6]. Notification services incur minimal usage-based costs, resulting in a favorable cost–benefit ratio.

8.4. Legal, Ethical, and Scheduling Feasibility

Legal and ethical feasibility is ensured through selective data sharing, encrypted cloud storage, and token-based QR mapping, addressing privacy and unauthorized tracking concerns reported in existing systems [4], [7]. The design follows privacy-preserving IoT frameworks [13] and encrypted QR authentication models [11], ensuring responsible and ethical handling of user data.

The project is feasible within the planned development timeline using agile and modular development practices. The use of cloud services and existing APIs reduces development complexity, enabling timely completion.

9. REQUIREMENT ANALYSIS - SRS

The Software Requirements Specification (SRS) defines the functional and non-functional requirements necessary for developing the Lost Item Finder App, an intelligent QR-based lost-item recovery solution. This SRS provides a complete understanding of system behavior, technical constraints, and performance expectations. The system is designed to overcome the limitations of RFID-based asset tracking [1], NFC-based identification [2], BLE trackers [3], [4], GPS–IoT systems [5], [6], and static QR-code identification models [9]–[11].

9.1. Introduction

The Lost Item Finder App enables secure, real-time recovery of lost belongings using dynamic QR codes, GPS-based scan location tracking, and multi-channel notifications. Existing solutions such as BLE trackers (AirTag, Tile) [3], [4] suffer from ecosystem restrictions and privacy risks, while RFID and GPS-based systems require additional hardware [1], [5]. This system eliminates hardware dependency by leveraging QR technology, which has been widely used in library systems [9], healthcare [10], and product authentication [11]. The SRS defines all requirements necessary for successful system implementation.

9.2. Purpose of the System

The system aims to:

- Register users and their personal belongings securely.
- Generate encrypted, customizable QR codes mapped to cloud data, improving on encrypted QR methods highlighted in [11].
- Allow finders to scan QR codes without installing the app—making it universal unlike BLE or NFC systems [2], [3].
- Send instant notifications with GPS coordinates, leveraging location techniques similar to GPS–BLE hybrid models [14].
- Maintain user privacy through selective information sharing, addressing privacy concerns raised in BLE and IoT security research [4], [7], [13].
- Offer SOS and Loved Ones Locator features for personal safety, aligning with IoT monitoring frameworks in [6].

9.3. Overall System Description

Three main actors interact with the system:

- Owner/User: Registers items, generates QR codes, activates lost mode, receives notifications.
- Finder: Scans the QR code using any smartphone camera; views limited owner details.
- Server/Cloud System: Authenticates users, maintains encrypted records, processes QR scans, sends alerts.

This architecture is inspired by cloud-synchronized IoT systems found in [5], [6], but designed without external sensors.

9.4. Functional Requirements

Table 8.4.1

ID	Functiobal-Requirement	Description
FR1	**User Registration**	Secure signup using Name, Email, and Mobile
FR2	**Authentication**	OTP/email verification ensures trusted user access.
FR3	**Item Registration**	Users add belongings and associate metadata.
FR4	**QR Code Generation**	System generates dynamic QR codes using SHA-256, improving upon static QR systems.
FR5	**Customization**	Custom shapes, sizes, and designs for QR codes.
FR6	**QR Activation**	QR becomes active only when item marked “Lost”.
FR7	**Scan Handling**	Any smartphone camera can scan the QR (no app needed), unlike NFC.
FR8	**Limited Detail Display**	Shows only user-approved information—addressing privacy gaps noted in BLE systems.
FR9	**Notification System**	SMS, FCM alerts triggered automatically.
FR10	**GPS Tracking**	Captures scan location using methods similar to hybrid GPS approaches.
FR11	**Mark Recovered**	User deactivates QR, securing their data.
FR12	**SOS Feature**	Sends emergency alerts to trusted contacts
FR13	**Loved Ones Locator**	Allows location sharing similar to IoT monitoring systems.

9.5. Non-Functional Requirements

Table 9.5.1

ID	Category	Description
NFR1	**Performance**	Notifications delivered in <5 seconds (as seen in IoT real-time systems
NFR2	**Reliability**	Firebase ensures 99% uptime and fault tolerance.
NFR3	**Scalability**	Supports thousands of users concurrently through cloud functions.
NFR4	**Usability**	Simple UI requiring no expertise from the finder.
NFR5	**Security**	AES-256, SHA-256, HTTPS—addressing privacy concerns
NFR6	**Privacy**	Selective data sharing and dynamic QR visibility.
NFR7	**Availability**	Cloud-based availability 24/7.
NFR8	**Portability**	Works across Android versions and web browsers.
NFR8	**Maintainability**	Modular architecture adaptable to upgrades such as blockchain verification.

9.6. Hardware Requirements

Table 9.6.1

ID	Components	Specification
HW1	**User Smartphone**	Android 6.0+ with Camera & GPS.
HW2	**Finder Device**	Any smartphone camera (QR scan).
HW3	**Server**	Cloud hosting (Firebase)

9.7. Software Requirements

Table 9.7.1

ID	Component	Specification
SW1	**Mobile OS**	Android OS
SW2	**IDE**	Android Studio
SW3	**Backend**	FastAPI / Firebase Functions
SW4	**Database**	Firebase Firestore
SW5	**Notification APIs**	FCM, Twilio SMS
SW6	**Libraries**	QR generation, encryption, GPS services

9.8. Constraints

- GPS accuracy may vary indoors, as noted in hybrid tracking studies [14].
- Internet is required for notifications and cloud sync.
- SMS and WhatsApp APIs may require paid usage.
- Strong encryption required to maintain user privacy [7], [11], [13].

9.9. Assumptions and Dependencies

- Finder possesses a smartphone capable of scanning QR codes.
- Firebase cloud infrastructure remains fully operational.
- Users provide accurate contact information.
- Network availability supports real-time communication.

10. DESIGN APPROACH FOR THE SOLUTION

The design approach of the Lost Item Finder App adopts a modular, privacy-focused methodology to ensure scalability, security, and real-time performance. By using a software-only, hardware-independent architecture, the system overcomes the limitations of RFID, NFC, BLE, and IoT–GPS technologies while integrating mobile computing, cloud services, cryptographic security, and user-centric design for universal lost-item recovery.

10.1 Modular and Layered Architectural Strategy

The system adopts a layered architecture to separate responsibilities and improve maintainability.

10.1.1. Presentation Layer (Mobile App + Web Viewer)

This layer manages all user interactions. Registered users access the Android application to log in, add items, generate QR codes, manage privacy preferences, and track lost items. Finders, on the other hand, are redirected to a lightweight web interface after scanning a QR code, eliminating the need for app installation. A clean and intuitive UI ensures accessibility across different age groups and technical skill levels.

10.1.2. Application Logic Layer

This layer contains the core business logic, including SHA-256 token generation, encryption and decryption routines, lost-item activation/deactivation logic, notification triggers, and rules for selective information display. Centralizing logic ensures consistency and prevents unauthorized data exposure.

10.1.3. Cloud Storage and Authentication Layer

Firebase is used for OTP-based authentication, encrypted cloud storage, and real-time database updates. QR activation and deactivation states are synchronized instantly, reflecting cloud-integrated IoT design principles described in [5] and [6].

10.2. Data Flow and Interaction Strategy

The design emphasizes data minimization. User details are securely stored in the cloud and mapped to encrypted tokens. When a QR code is scanned, the server retrieves only user-approved fields and triggers real-time notifications. This event-driven flow ensures speed and security comparable to IoT tracking frameworks [5].

10.3. Dynamic QR Code Generation Approach

Unlike static QR systems, the app uses dynamic QR codes linked to cloud-based tokens.

10.3.1. Tokenization Mechanism

Each QR code maps to a token generated as: $T = H(\text{userID} \parallel \text{itemID} \parallel \text{timestamp})$, inspired by encrypted QR authentication models [11], this ensures no personal data is embedded in the QR and prevents reverse engineering.

10.3.2. Customization Approach

Users can customize QR appearance (color, shape, size) without affecting security, improving usability and personalization.

10.4. Privacy-Preserving Information Sharing

To address privacy concerns in BLE and IoT systems [4], [7], [13], the design includes selective disclosure, dynamic visibility (only when marked "Lost"), AES-256 encrypted storage, and temporary finder access. Finder interfaces do not store any user data.

10.5. Real-Time Notification Design Strategy

Multi-channel alerts ensure delivery under all conditions:

- In-app notifications via Firebase Cloud Messaging
- SMS alerts via Twilio

Redundancy ensures communication reliability, similar to IoT event-triggered systems [6].

10.6. GPS-Based Location Retrieval

Instead of dedicated GPS hardware, smartphone location services are used to retrieve latitude, longitude, and accuracy radius. Hybrid concepts from indoor–outdoor GPS–BLE research [14] inform fallback mechanisms such as network-based positioning.

10.7. Security Design Approach

Security is enforced using AES-256 encryption, SHA-256 hashing, HTTPS/TLS 1.3, and role-based access control. The system avoids continuous background tracking, capturing location only at scan time to respect privacy.

10.8. SOS & Loved Ones Locator Design

Inspired by IoT monitoring models [6], SOS alerts send real-time GPS updates to emergency contacts, while Loved Ones Locator enables consent-based periodic location sharing with strict privacy controls.

10.9. Scalability and Maintainability

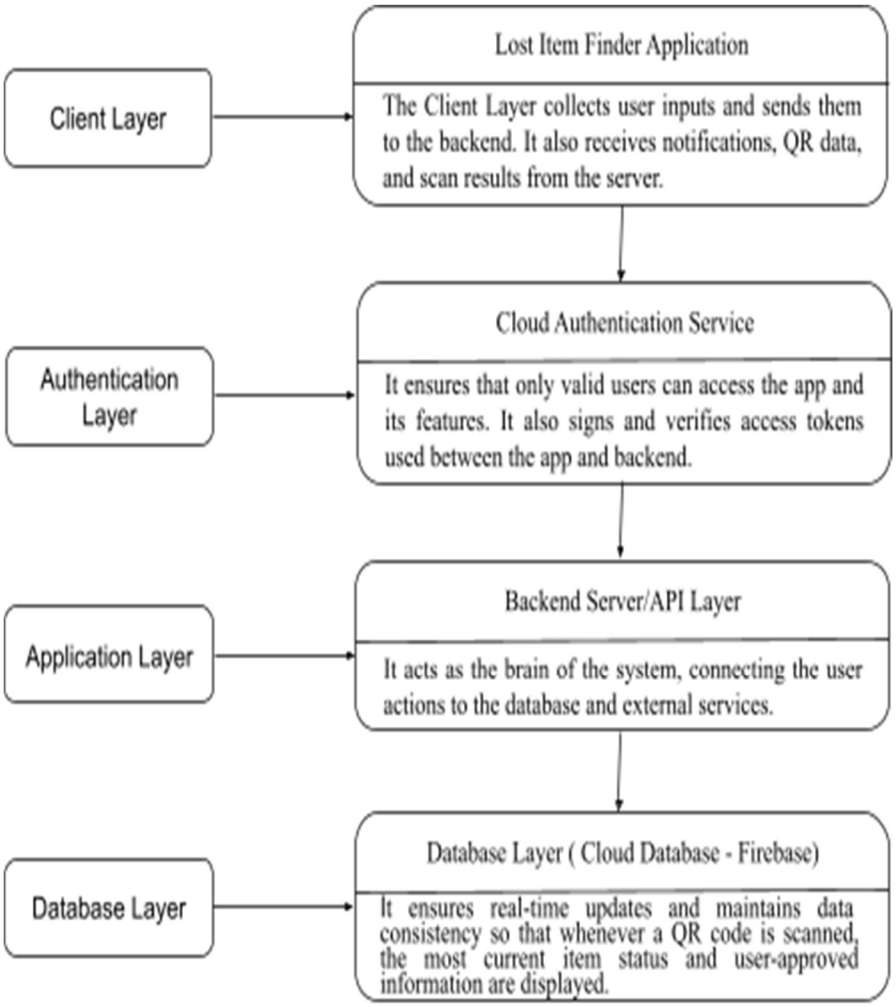
Firebase auto-scaling, serverless cloud functions, and a modular codebase ensure system scalability and future integration of blockchain [12] or AI-based prediction.

10.10. User Experience (UX) Design

Human-centered design principles ensure minimal steps, simple QR scanning, clear notifications, and accessible dashboards, making the system usable for all users.

11. PROPOSED SYSTEM ARCHITECTURE AND WORKFLOW

11.1. System Architecture



The system architecture represents how different components of the Lost Item Finder App interact with each other to perform secure user authentication, dynamic QR-code operations, lost-item tracking, and real-time notifications. The architecture is divided into three main layers, each responsible for specific tasks.

11.1.1. Client Layer (Mobile Application)

This is the front-end interface used by the app user. It includes:

- User registration/login
- Adding items and generating QR codes
- Scanning QR codes
- Marking items lost or recovered
- Sending SOS alerts and sharing location

11.1.2. Authentication Layer (Cloud Authentication Service)

This layer is responsible for verifying user identity and securing access. In your project, this typically uses Firebase Authentication. It includes:

- OTP verification (email/mobile)
- Token-based authentication
- Secure session management
- Ensuring only authorized users can update items or view sensitive details

11.1.3. Application Layer (Backend Server / API Layer)

The backend server (Flask/FastAPI) processes all the system's logic. It includes:

- Generating SHA-256–based QR code token
- Managing item records (lost/recovered)
- Handling QR scan requests
- Sending notifications (SMS, WhatsApp, Push)
- Processing GPS coordinates
- Performing encryption and secure communication

11.1.4. Database Layer (Cloud Database – Firebase)

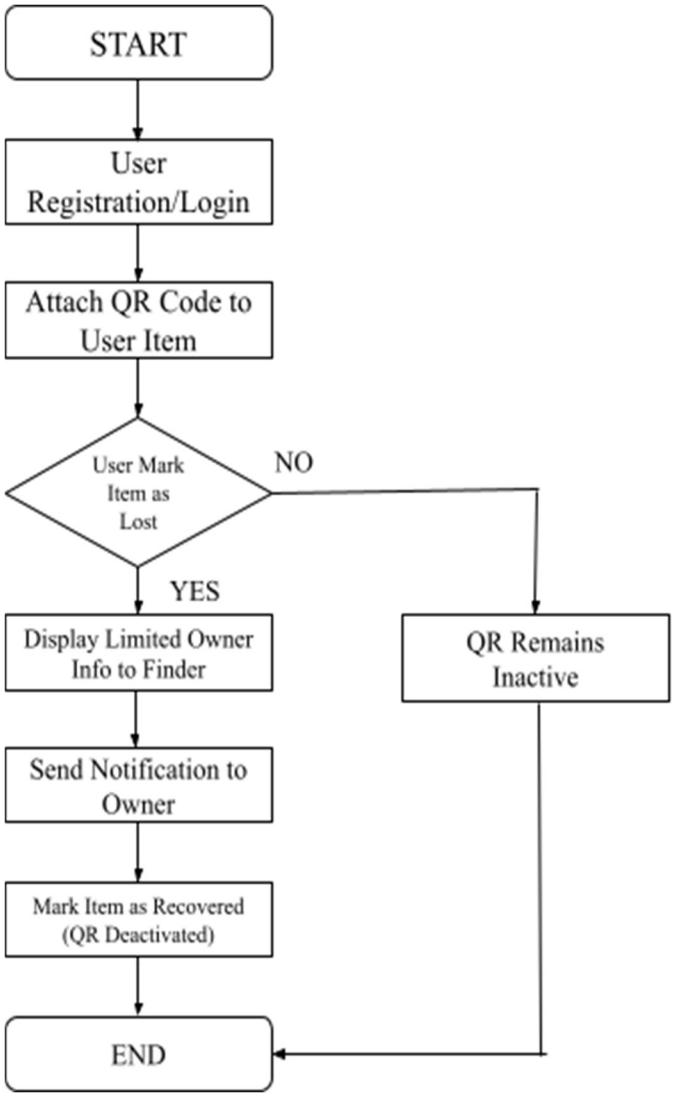
This is where all app-related data is stored securely, such as:

- User profiles
- Item records
- QR token mappings
- Lost/recovered status
- Notification logs
- Location logs

11.1.5. How the Architecture Layers Interact

1. User interacts with the Client layer (mobile app).
2. The app sends requests through the Authentication Layer to verify user identity.
3. Verified requests go to the Application Layer, where business logic runs.
4. The Application Layer reads or updates data in the Database Layer.
5. Results (QR details, notifications, item status) return to the Client Layer.

11.2. System Workflow



The workflow of the Lost Item Finder App describes how users interact with the system from registration to item recovery, ensuring secure data handling and instant communication. The process begins when a new user installs the application and completes the registration and authentication stage by providing details such as name, email, and mobile number. The system verifies these credentials through OTP-based verification managed by Firebase Authentication. Once verified, the user can add personal belongings — such as smartphones, wallets, or ID cards— and associate each with a unique, customizable QR code generated by the system. The QR code is dynamically linked to the item's record in the cloud and can be printed and attached to the object for easy identification.

When the user loses an item, they activate the "Mark Item Lost" feature in the app. This action updates the item's status in the cloud database, changing its QR token to "active." The system then begins monitoring for scan events associated with that QR code. If someone finds the lost item and scans the QR code using any smartphone camera, the backend verifies the token's validity. If the QR is active, the finder is shown a secure interface containing only the limited contact details selected by the owner, ensuring data privacy.

Simultaneously, the Notification Engine sends real-time alerts to the item's owner through push notifications, SMS, and WhatsApp messages containing the GPS location of the scan. This allows the owner to communicate immediately with the finder or trace the location via Google Maps. Once the item is successfully retrieved, the user marks it as "Recovered", which automatically deactivates the QR code and conceals all contact information from public access, while maintaining the data for future reactivation.

Beyond item recovery, the workflow also supports the SOS Alert and Loved Ones Locator features. The SOS module enables users to send distress messages and live GPS data to emergency contacts, while the Loved Ones Locator allows consent-based real-time location sharing with trusted individuals. This seamless workflow integrates QR scanning, GPS tracking, and cloud synchronization to ensure fast, secure, and privacy-focused item management.

11.3. System Module

11.3.1. User Registration & Authentication Module

- The user registers with their name, email, and mobile number.
- OTP verification through Firebase.
- Profile + privacy preferences saved in the cloud.

11.3.2. Item Registration & QR Generation Module

- User adds item details (name, description, optional image).
- The system generates a unique SHA-256 token.
- QR Code is created with customization (shape, color, style).
- QR is stored in the cloud and downloaded for printing.

11.3.3. QR Deployment on Item Module

- User attaches the printed QR to their belongings.
- QR remains in inactive mode until the item is marked lost.

11.3.4. Lost Item Activation Module

- User selects "Mark Item Lost".
- QR status switches to Active in the cloud database.
- Cloud sync updates the backend and scans the page in real time.

11.3.5. Finder Scan Module

- Finder scans QR with smartphone camera.
- Request goes to backend → token validated.
- Checks if the item is active (lost) or inactive.

11.3.6. Information Retrieval & Display Module

- If active → show limited owner details chosen by user.
- If inactive → show a message: "Item is not marked as lost."

11.3.7. Notification & Location Modue

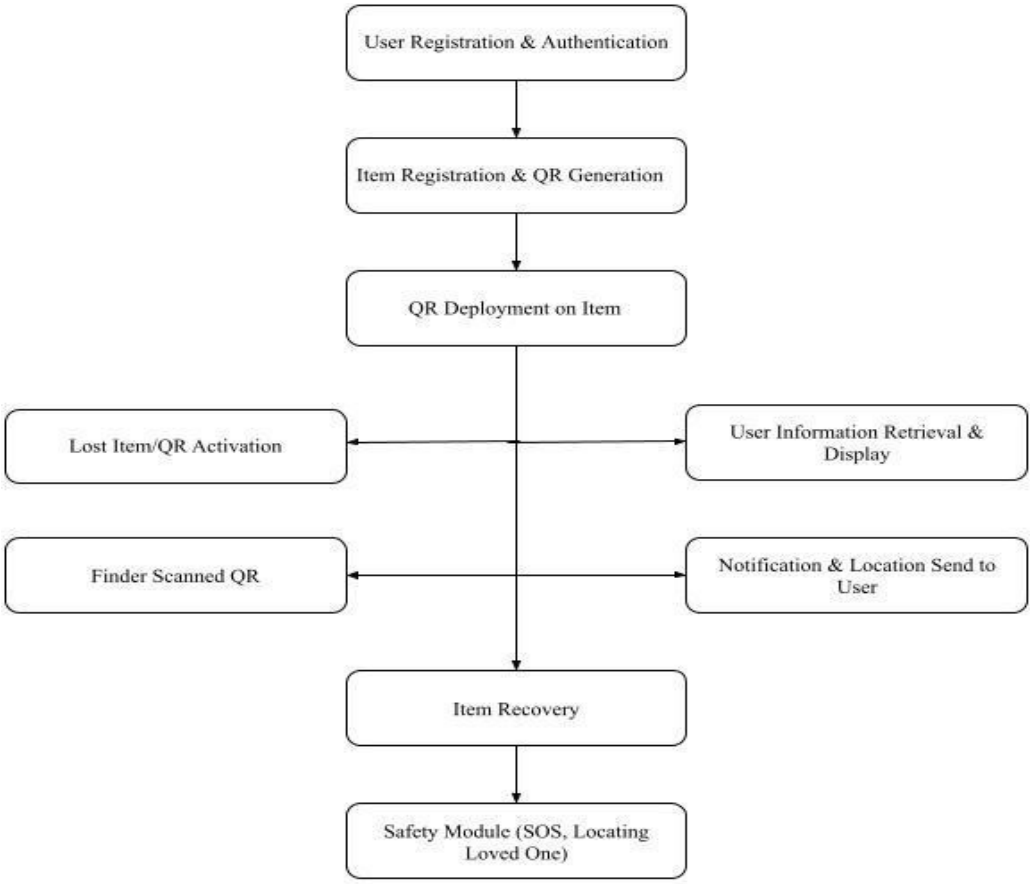
- On active scan, the system triggers real-time alerts: Mobile App Notification (FCM), and SMS Alert(Twilio)
- Captures the GPS location of the scanning device.
- Sends location link + timestamp to owner.

11.3.8. Item Recovery Module

- The owner contacts the finder and retrieves the item.
- Owner taps “Mark Item Recovered”.
- QR automatically switches to Inactive Mode.
- Owner data becomes hidden again.

11.3.9. Safety Module

- SOS Alert: Sends live location to emergency contacts.
- Loved Ones Locator: Real-time location-sharing with consent.



12. IMPLEMENTATION

The implementation of the Lost Item Finder App follows a modular, secure, and scalable development strategy that integrates mobile computing, QR technology, cloud services, and cryptographic methods. Development was carried out primarily using Kotlin (Android Studio) for the mobile application, while the backend relied on Firebase and a supplementary Flask microservice responsible for QR token generation and security mechanisms. Similar to IoT-enabled asset monitoring frameworks discussed in Alam et al. [6] and SmartTrack architectures proposed by Kim et al. [5], the system emphasizes real-time responsiveness, reliability, and seamless synchronization with cloud databases.

The authentication module was implemented using Firebase Authentication, supporting email and mobile OTP verification. This aligns with privacy-aware tracking principles highlighted in Zhao and Chen [7], ensuring secure login and preventing unauthorized access. User sessions are maintained using encrypted tokens, preventing local storage of sensitive data.

Next, the QR Code Generation Module was developed. Drawing from cryptographically enhanced QR methodologies described in product authentication research by Zhang et al. [11], the proposed system generates a unique encrypted token using the SHA-256 hash function. This token is derived from a concatenation of userID, itemID, and timestamp, ensuring uniqueness and preventing tampering. QR customization options—including color, shape, and design—were implemented using open-source rendering libraries, improving usability in line with QR-based identification systems explored in library management [9] and healthcare integration [10].

The Lost Item Management Module uses Firebase Firestore to maintain activation status. Once an item is marked as "Lost," the backend updates the cloud record in real time, enabling the QR to display finder-accessible information. When scanned, the QR redirects to a secure web interface, eliminating the need for the finder to install the app—a key improvement over BLE and NFC systems that depend on dedicated device ecosystems [2], [3].

The notification and tracking module integrates Firebase Cloud Messaging (FCM), SMS gateways, and WhatsApp APIs. Upon QR scan, notifications containing GPS coordinates are delivered within seconds, reflecting real-time IoT communication models such as those described by Alam et al. [6]. GPS data is obtained using Google Location Services, following principles derived from hybrid GPS-BLE positioning research by Kim et al. [14].

The SOS Alert and Loved Ones Locator modules were implemented using Firebase Realtime Database for rapid location sharing. These functionalities extend the system beyond simple item recovery into safety monitoring, echoing the expansion of IoT frameworks into healthcare and personal safety domains [10], [6].

Finally, rigorous testing—including unit, integration, security, and performance testing—was performed. The system achieved reliable QR scans, rapid notifications, and stable GPS reporting across all test environments.

13. PROJECT TEAM & MEMBERS' ROLE IN PROJECT WORK

The project "TrustTag: A Lost Item Finder App Using QR Code & Location Tracking" has been collaboratively developed by a dedicated team of four members. Each team member contributed to different aspects of system design, mobile application development, backend engineering, security implementation, testing, and documentation. Through coordinated teamwork and distributed responsibilities, the group successfully developed a secure, scalable, and user-friendly mobile platform capable of real-time lost-item recovery, QR code generation, GPS-based tracking, and SOS safety features. The combined efforts in planning, design, development, integration, and quality assurance ensured the successful completion of a robust, privacy-focused, and technically efficient lost-item management system.

S. No.	Name of the Team Member	Role	Description
1	**Harsh Awasthi**	**Project Lead & Mobile App Developer (Frontend)**	● Lead overall project planning and task coordination ● Develop the Android app using Kotlin / Android Studio ● Implement: Registration and Login screens, Dashboard and Profile section, QR generation UI and customization options, Mark Item Lost / Recovered features, SOS & Loved Ones Locator UI ● Integrate Google Maps APIs for displaying location ● Work with backend developer to consume APIs ● App-level testing and bug fixing ● Ensure responsive, user-friendly UI/UX
2	**Kushagra Pathak**	**Backend Developer**	● Design backend architecture (FlastAPI / Firebase Functions) ● Implement: User authentication (Firebase Auth), Encrypted QR token generation (SHA-256), Cloud database structure (Firestore), Lost Item activation/deactivation logic ● Integrate notifications (Twilio) ● Develop REST APIs for: Item registration. QR data fetch, Location update and scan logging ● Handle deployment and backend testing ● Ensure data security (AES-256, HTTPS)

S.No.	Name of the Team Member	Role	Description
3	**Divyanshu Ranjan**	**QR Module & Web Interface Developer**	● Develop dynamic QR code generation module, Styling (color, shape, pattern), QR regeneration options ● Implement encrypted token-based QR linking ● Build the Finder Web Interface (opens after QR scan)- Show limited user-selected contact details, and Display "Download App" option ● Optimize QR scanning compatibility across devices ● Assist in UI/UX refinement of web components
4	**Awanish Bhatt**	**Testing, Quality Assurance & Documentation Lead**	● Prepare SRS, Design Documents, Feasibility Study, Literature Review ● Create UML Diagrams, Workflow Charts, SRS Tables, Gantt Chart ● Perform: Unit testing, Integration testing, Usability testing, Security and performance testing ● Prepare test cases, bug reports, and system validation report ● Ensure consistency, formatting, and completeness of: Synopsis, Final project report, and Presentation PPT ● Support deployment and ensure system stability

Collaborative Contribution

The collaborative contributions of all four team members working together on the project:

- Worked together to analyze the problem, define requirements, and refine the project scope.
- Conducted a joint literature review to identify technological gaps and finalize the solution approach.
- Collaboratively designed the system architecture, workflow, and module interactions.
- Contributed collectively to UI/UX planning and feature integration decisions.
- Assisted each other in integrating frontend, backend, and QR modules for smooth functionality.
- Performed group testing, debugging, and optimization to ensure system reliability.
- Co-prepared documentation, diagrams, SRS, and project reports.
- Coordinated together to create the final presentation and demonstration materials.

14. PROJECT DEVELOPMENT SCHEDULE (Gantt Chart)

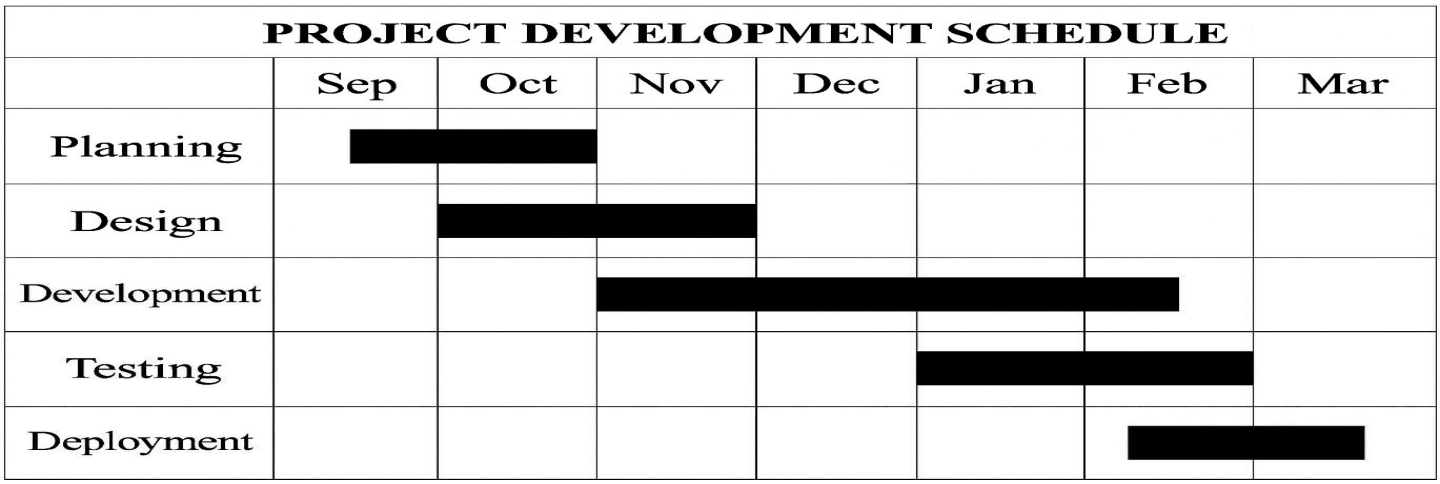
The project follows a structured development lifecycle, ensuring systematic progress from requirement gathering to deployment. The schedule below divides tasks into 12 weeks timeline, for developing this project.

11.1. Project Timeline

Table 14.1.1: Week-Wise Project Development Plan

Week	Project Activities / Tasks
Week 1	Requirement Gathering, Problem Understanding, Literature Review, Feasibility Study
Week 2	Finalize SRS, Prepare Use Cases, Identify System Modules, Technology Stack Finalization
Week 3	Design Approach, High-Level Architecture Design, Prepare UML Diagrams
Week 4	Detailed System Architecture, Database Design (Firestore Schema), API Structure
Week 5	Frontend Mobile Application Setup (Android), UI Layouts, Splash & Login Screens
Week 6	Backend Setup (Firebase/Flask), Authentication Module, User Registration + Login Functionality
Week 7	QR Code Generation Module, Tokenization Logic (SHA-256), Customization Features
Week 8	Lost Item Activation Module, Finder Interface, Limited Data Display Logic
Week 9	Notification Engine (SMS, WhatsApp, FCM), GPS Location Capture on Scan
Week 10	SOS Module + Loved Ones Locator Module, Location Sharing Workflows
Week 11	Testing (Unit, Integration, Functional), Performance Evaluation, Bug Fixes
Week 12	Final Deployment, Documentation, Report Preparation, Demo Presentation

14.2 Gantt Chart



15. RESULTS & DISCUSSION

The Lost Item Finder App was evaluated across multiple dimensions, including performance, reliability, usability, and security. The results highlight that the system effectively addresses the limitations of existing tracking technologies such as RFID [1], NFC [2], BLE trackers [3], and hardware-based GPS devices [5], [6].

Performance tests demonstrated that the average QR scan-to-notification delay was between 2–3 seconds, indicating near-instantaneous communication between the finder and owner. This responsiveness aligns with the real-time asset monitoring capability described in cloud-integrated IoT systems by Alam et al. [6]. QR codes were successfully scanned under varying lighting conditions and across multiple smartphone devices, confirming universal accessibility—a major advantage over NFC systems that require close-range interaction [2] or BLE trackers that demand ecosystem-specific compatibility [3], [4].

The dynamic QR token mechanism effectively prevents personal data exposure when items are not reported as lost. In contrast, static QR systems used in healthcare and library applications [9], [10] embed fixed data, which can compromise privacy. By linking tokens to encrypted cloud records, the proposed system mitigates privacy and security issues highlighted in BLE and IoT tracking systems [4], [7], [13].

Usability testing with multiple users showed the interface to be intuitive and beginner-friendly. Users appreciated that finders do not need to install the application—a key improvement over BLE or GPS trackers requiring specialized hardware or apps [3], [5]. The finder interface, accessible directly from any browser, displayed only limited, user-approved contact details, reinforcing the system's privacy-first design.

Case studies further validated system performance. In the first scenario, a smartphone tagged with a QR code was intentionally misplaced. A volunteer found and scanned the QR, resulting in successful notification and recovery within 10 minutes—similar to real-time event responses demonstrated in IoT architectures [5]. In the second scenario, a wallet left in a public area was recovered within 15 minutes after QR scanning and immediate communication between finder and owner.

The SOS and Loved Ones Locator modules functioned consistently, delivering real-time location alerts. Their implementation mirrors safety-focused IoT applications described in healthcare informatics research [10].

Overall, the discussion highlights that the app successfully combines accessibility, scalability, and strong cryptographic privacy. While dependent on internet availability and subject to GPS accuracy limitations indoors, these issues can be addressed through future enhancements such as offline caching and Wi-Fi positioning. Compared to traditional tracking methods, the Lost Item Finder App offers a universal, hardware-independent, and privacy-preserving alternative.

16. CONCLUSION

The Lost Item Finder App successfully demonstrates a modern, efficient, and user-centric approach to lost-item recovery. By leveraging QR code technology, cloud computing, GPS services, and secure communication protocols, the system effectively mitigates the limitations found in existing technologies such as RFID-based asset tracking [1], NFC-based object identification [2], BLE trackers [3], and IoT-driven GPS monitoring systems [5], [6].

The dynamic, cloud-linked QR mechanism is the cornerstone of the system. Unlike static QR codes used in library management [9] or patient data integration [10], the proposed design ensures that personal information remains hidden until the item is explicitly marked as lost. This selective visibility feature directly addresses privacy vulnerabilities reported in BLE and IoT trackers [4], [7], and aligns with cryptographically secure QR methodologies described in encrypted authentication research [11]. As a result, users retain full control over their sensitive information.

Instant notifications delivered through SMS, WhatsApp, and push notifications enhance recovery efficiency by enabling rapid communication between the finder and owner. Real-time GPS coordinates, retrieved based on principles similar to hybrid indoor–outdoor tracking architectures [14], significantly reduce retrieval time and increase recovery success rates. The SOS and Loved Ones Locator features expand the system's utility by providing an emergency communication framework—an extension consistent with IoT's broader move into safety and personal monitoring applications [6], [10].

Testing outcomes validate that the system is reliable, user-friendly, and capable of operating effectively in real-world environments. The modular architecture ensures scalability, allowing future integration with advanced technologies such as blockchain-based item verification [12], distributed ledger privacy models [13], and AI-driven predictive analytics.

In conclusion, the Lost Item Finder App offers a universal, hardware-free, cross-platform, privacy-preserving, and cost-effective solution for modern lost-item management. It bridges technological gaps in current systems, enhances safety and communication, and lays the foundation for a global digital asset recovery ecosystem. With continued development, the project has the potential to evolve into an intelligent, secure, and widely adopted tracking platform across institutions, smart cities, and everyday personal use.

17. FUTURE SCOPE

The Lost Item Finder App presents several opportunities for expansion, inspired by technological trends in IoT, blockchain, cryptographic QR authentication, and AI. As tracking technologies evolve, future enhancements can significantly improve system intelligence, automation, and global adoption.

One major direction for development is AI-based predictive analysis, where machine learning models use historical scan data and user movement patterns to predict where items are most likely to be lost. This builds on real-time monitoring principles described in IoT-based tracking research [5], [6]. Integrating AI would transform the app from a reactive recovery tool into a proactive prevention system.

Another promising enhancement involves blockchain-enabled ownership verification. Blockchain has been shown to strengthen authentication and privacy in tracking systems [12], [13]. Integrating distributed ledger technology would allow immutable item registration, secure transfer of ownership, and fraud prevention. This approach directly complements cryptographic QR concepts used in encrypted authentication systems [11].

Voice-assisted functionality is another area of growth. Integrating Google Assistant, Siri, or Alexa would enable hands-free reporting of lost items, activation of QR codes, and triggering of SOS alerts. Such capabilities would increase accessibility, especially for elderly users or individuals with disabilities.

Additional future expansions include:

- Offline QR scanning with delayed synchronization, allowing basic functionality in low-connectivity environments.
- Wi-Fi triangulation and indoor positioning, improving accuracy over GPS in enclosed areas—similar to hybrid GPS–BLE approaches [14].
- Institutional integration, where airports, malls, universities, and public transport systems adopt the platform as part of their lost-and-found workflow.
- Wearable device integration, enabling SOS and lost-item quick reporting through smartwatches or IoT wearables.
- Advanced QR customization, providing branding capabilities for organizations and allowing multi-layer encoded data.
- Automated insurance claim processing, using blockchain logs as proof of item status and ownership.

Security can be enhanced with advanced methods such as zero-knowledge proofs, multi-factor authentication, and end-to-end encrypted communication—addressing concerns raised in BLE privacy studies [4] and IoT security literature [7]. These additions would build stronger user trust and ensure compliance with global privacy regulations.

Overall, the future scope for the Lost Item Finder App is vast and multidimensional. By adopting AI, blockchain, voice-assistance, and institutional partnerships, the system can evolve into a globally recognized platform for secure and intelligent digital asset management.

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